

C2.1 How have our ideas about atoms developed over time?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
The modern model of the atom developed over time. Stages in the development of the model included ideas by the ancient Greeks (4 element ideas), Dalton (first particle model), Thomson ('plum pudding' model), Rutherford (idea of atomic nucleus) and Bohr (shells of electrons). Models were rejected, modified and extended as new evidence became available. The development of the atomic model involved scientists suggesting explanations, making and checking predictions based on their explanations, and building on each other's work (1aS3). The Periodic Table can be used to find the atomic number and relative atomic mass of an atom of an element, and then work out the numbers of protons, neutrons and electrons. The number of electrons in each shell can be represented by simple conventions such as dots in circles or as a set of numbers (for example, sodium as 2.8.1). Atoms are small – about 10^{-10} m across, and the nucleus is at the centre, about a hundred-thousandth of the diameter of the atom. Molecules are larger, containing from two to hundreds of atoms. Objects that can be seen with the naked eye contain millions of atoms.	1. describe how and why the atomic model has changed over time to include the main ideas of Dalton, Thomson, Rutherford and Bohr
	2. describe the atom as a positively charged nucleus surrounded by negatively charged electrons, with the nuclear radius much smaller than that of the atom and with most of the mass in the nucleus
	3. recall relative charges and approximate relative masses of protons, neutrons and electrons
	4. estimate the size and scale of atoms relative to other particles M1d
	5. recall the typical size (order of magnitude) of atoms and small molecules
	6. relate size and scale of atoms to objects in the physical world M1d
	7. calculate numbers of protons, neutrons and electrons in atoms, given atomic number and mass number of isotopes or by extracting data from the Periodic Table M1a

Linked learning opportunities

Specification links:

- atoms and radiation (P5.1)

Ideas about Science:

- understanding how scientific explanations and models develop in the context of changing ideas about the atomic model (1aS3)